Knowledge-Based Visual Question Answering

Unimodal Baselines and CLIP Fusion on A-OKVQA

Deep Learning and Its Applications (CO3133)

Semester 261 – Class CC01 – Group G-02

Assignment 3: Multimodal Deep Learning

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Problem Statement & Research Objectives

Problem

Given an image, a question and 4 choices, pick the correct choice. The questions need commonsense or world knowledge.

$$s_j = f_\theta(\text{image}, \text{question}, \text{choice}_j), \qquad \hat{y} = \arg\max_j s_j$$

Research objectives

- Build modality-A (text) and modality-B (image) baselines, a simple fusion and an improved fusion.
- Measure how much each modality contributes, and when they support or conflict.
- Ablate the fusion level, the fusion operation and fine-tuning of the encoder.

Challenge

Answers need both what is visible and outside knowledge; only 15k training questions.

Dataset & Data Splitting

Dataset: A-OKVQA (COCO images)

- Hugging Face, revision `d1b0efa`, Apache 2.0.
- 18,201 labelled questions on 17,661 images; 4 choices, 10 free-form answers, 3 rationales each.
- Official test split has no labels → not used.

Split	Questions	Images	Purpose
Train	15,346	14,885	training
Validation	1,710	1,654	checkpoint
Test	1,145	1,122	final, once

Split **by image** (MD5 of the bytes), seed 42; 540 images with 2 questions stay together.



*Figure 1: Random training pairs; * marks the correct choice.*

Exploratory Data Analysis: Pairs, Labels & Text Priors

0 missing images or questions

no missing-modality handling needed; 44 questions with an empty or repeated choice are kept.

24–26% per letter

A / B / C / D are balanced, so letter bias cannot help.

32.4% low agreement

of questions: the top free-form answer has ≤ 3 of 10 workers.

Text-only rules (validation)

Rule	Accuracy
Chance	25.0%
Most frequent letter	26.0%
Answer-text prior	31.1%
Longest choice	28.5%

$\rightarrow$ text alone barely beats chance: the image should matter

Methodology: Pipeline

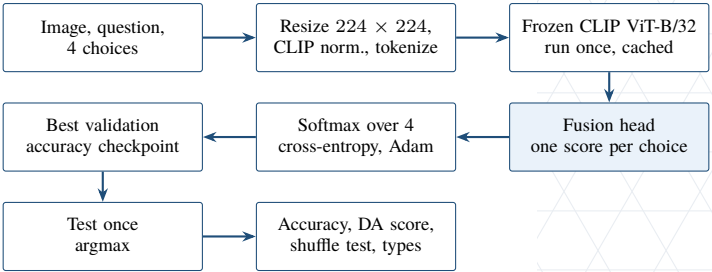


Figure 2: Raw paired data, preprocessing, encoders, fusion, loss, optimisation, prediction, evaluation.

Encoders (frozen, 151M params)

Image: global v (512) + 50 tokens (768). Text: q, c_j, "question answer: choice$_j$" vector and 51 tokens.

Cost of caching

99 s for 17,661 images, 72 s for 163,809 texts on one A40.

→ **every head trains in seconds**

Methodology: Models & Fusion

Model	Fusion stage	MLP input for choice j	Trainable
Zero-shot CLIP	none	$\cos(v, g_j)$, no training	0
TextOnly (A)	none	$[q;\ c_j;\ q \odot c_j]$	0.79M
ImageOnly (B)	none	$[v;\ c_j;\ v \odot c_j]$	0.79M
LateFusion	late	$[v;\ q;\ c_j;\ v \odot c_j;\ q \odot c_j]$	1.31M
CrossAttention	intermediate	LateFusion + region vector r_j	3.28M

```
CrossAttention.forward()
img = proj(norm(v_tokens))      # [B*4,50,512]
qry = norm(qc_tokens)           # [B*4,51,512]
att = attention(qry, img, img) # 8 heads
r = masked_mean(norm(qry+att)) # [B,4,512]
s = mlp([late inputs, r, r*g]) # [B,4]
```

- Loss: cross-entropy over the 4 choices (same for all models).
- Adam lr 10^{-3}, batch 64, 20 epochs, dropout 0.1, seeds 42–46.
- Ablation: train the last CLIP vision block (lr 10^{-5}, 7.1M params).

Methodology: Experimental Setup

E1: unimodal vs multimodal (M2)

TextOnly vs ImageOnly vs LateFusion.

E2: fusion level (M3)

LateFusion vs CrossAttention.

E3: fusion operation

LateFusion with and without the $\odot$ terms.

E4: freeze vs fine-tune

Last CLIP vision block frozen or trained.

E5: shuffled images

Test images replaced by images of other questions.

Fixed factors

Features, split, head size, optimizer, epochs, seeds.

Primary metric: test multiple-choice accuracy. **Secondary:** DA score $\min(n/3, 1)$, accuracy by question type, parameters, time.
A difference is supported when $|\bar{A} - \bar{B}| > \max(s_A, s_B)$, s = SD over 5 seeds (rule fixed in the proposal).

Results: Unimodal vs Multimodal

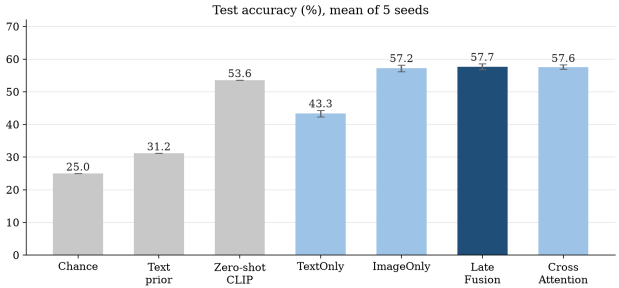


Figure 3: Test accuracy, mean ± SD of 5 seeds.

Key results

- Late fusion: 57.7 ± 0.9%, DA 0.470.
- +14.4 pt over text only, in 5/5 seeds.
- Only +0.5 pt over image only (below one SD).
- Zero-shot CLIP already 53.6% without training.

→ the image carries most of the answer

Results: Training Behaviour

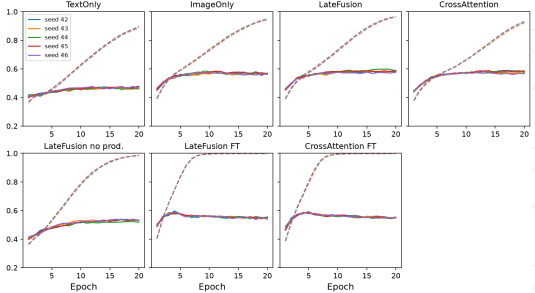


Figure 4: Validation (solid) and training (dashed) accuracy, one colour per seed.

Frozen features

Best epoch 9–20; train > 0.9, val ≈ 0.58.
$\rightarrow$ **the heads overfit, 20 epochs is enough**

Fine-tuned last block

Best epoch 3–5, then validation drops.
$\rightarrow$ **7.1M extra parameters memorise 15k questions**

Analysis: Does the Model Use the Image?

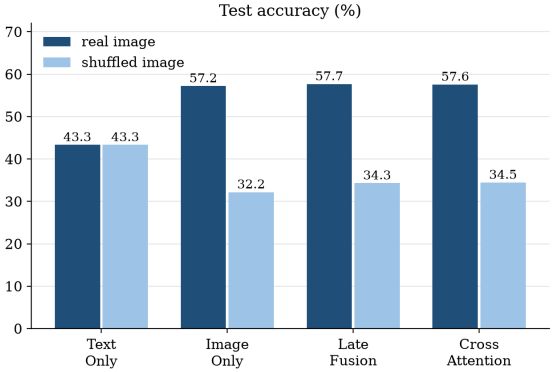


Figure 5: Test accuracy with the real image and with the image of another question.

−23 to −25 pt

for every image-based model when the image is shuffled.

Below text only

With a wrong image, fusion (34%) falls below text only (43%).
→ **the image is the main evidence, not a hint**

Analysis: Results by Question Type

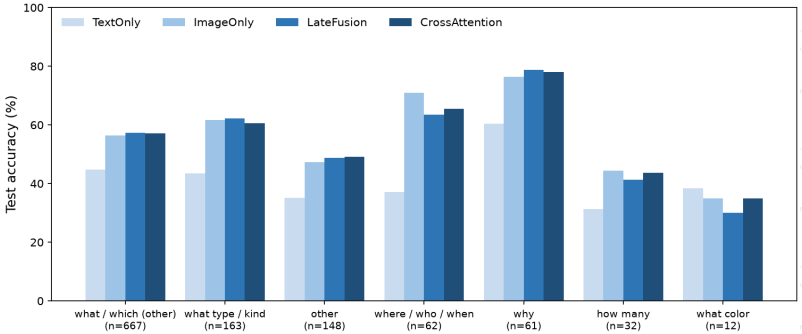


Figure 6: Test accuracy by question opening (mean over seeds).

Image helps most

where / who / when +26.5 pt, type / kind +18.8 pt.

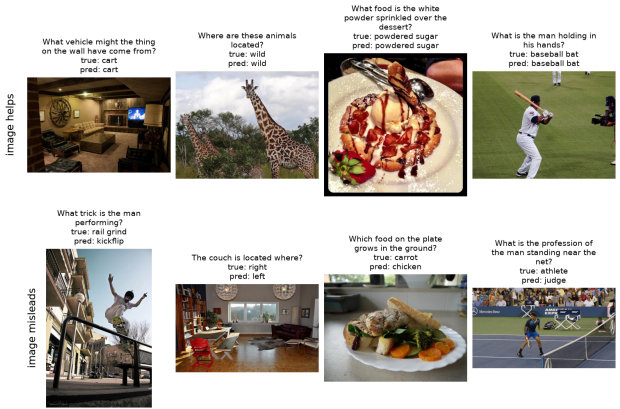
Why questions

Easiest (79%); the question adds +2.3 pt here.

Still hard

how many 41–44%, what color 30–35% (only 12 questions).

Analysis: Support vs Conflict Between Modalities



263 vs 101 (seed 42)

image helps vs image misleads; 2.6 : 1 over all seeds.

Support

The answer is clearly visible (baseball bat, powdered sugar).

Conflict

A plausible wrong answer is in the image: left vs right, chicken vs carrot.

→ **whole-image matching cannot reject it**

Ablation Study

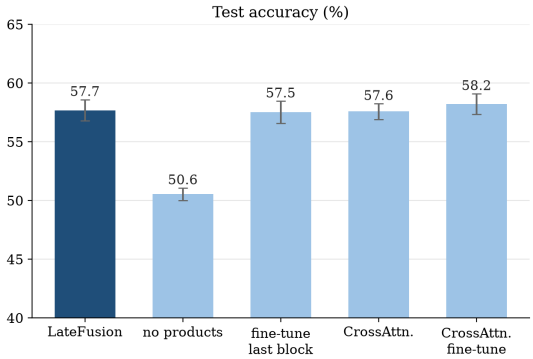


Figure 7: Test accuracy, mean ± SD of 5 seeds.

Change	Δ (pt)	Seeds
No products (E3)	−7.1	0/5
Cross-attention (E2)	−0.1	2/5
Fine-tune, late (E4)	−0.2	2/5
Fine-tune, cross (E4)	+0.6	3/5
Remove image	−14.4	0/5
Remove question	−0.5	1/5

→ **how vectors are combined matters more than where**

Analysis: Error Analysis

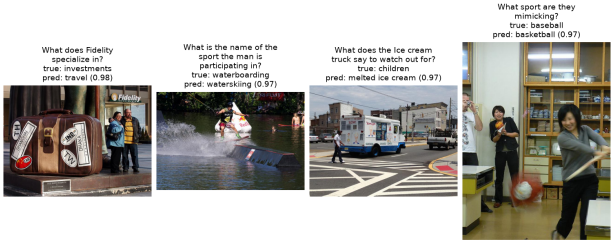


Figure 8: Most confident errors of the best model (CrossAttention + fine-tune, 58.2%).

Visible distractor

Suitcase → "travel" for Fidelity.
→ **region features**

Text in image & counting

Signs not read; how many: 52% errors.
→ **OCR, detector counts**

Ambiguous labels

Low agreement: 54.1% vs 59.8% accuracy.
→ **report DA score too**

Results: Compute Cost

Model	Trainable	Train (min)	Head (ms / q)	Test acc.
TextOnly	0.79M	0.10	0.002	43.3%
ImageOnly	0.79M	0.10	0.002	57.2%
LateFusion	1.31M	0.11	0.003	57.7%
CrossAttention	3.28M	0.92	0.056	57.6%
LateFusion + fine-tune	8.79M	0.95	0.053	57.5%
CrossAttention + fine-tune	10.77M	1.84	0.107	58.2%

- One-time CLIP caching on one A40: 5.6 ms per image, 0.44 ms per text; shared by all models.
- CrossAttention costs 9× LateFusion per epoch for the same accuracy.

→ **LateFusion is the best accuracy per cost**

Discussion & Conclusion

Modalities

Image +14.4 pt over text; question +0.5 pt over image.
→ **image–choice matching does most of the work**

Fusion operation

Element-wise products +7.1 pt, in 5/5 seeds.
→ **the biggest design effect**

Improved fusion

Cross-attention and fine-tuning stay within one SD.
→ **15k questions are few for region attention**

Interaction

Shuffled image: −23 to −25 pt; support : conflict = 2.6 : 1.
→ **a wrong image hurts more than no image**

Hard cases

Counting, colour, left / right, text in the image.

Limits

One frozen encoder; test = official validation (1,145); proposal without a formal decision when the runs started.

About 58% test accuracy: 33 points above chance and 4 above zero-shot CLIP; the image is the modality that matters.

Better fusion needs better region features or more data, not a bigger head.

AI Usage Disclosure

Claude Code, Claude Opus 5.5 (Võ Thị Như Quỳnh, 9 Nov 2026)

Dataset search and comparison; data loader, feature cache, models and training script; the 35 runs; `02_results.ipynb`; first drafts of the reports, these slides and the pages. Dataset choice and the decision to continue before approval: the group.

→ **every number comes from saved run outputs and executed notebooks**

Code, reports, slides and full disclosure (`AI_USAGE.md`): `https://quin210.github.io/dl-co3133/`